

FRACTIONS, DECIMALS AND PERCENTAGES, (COMPARING FRACTIONS, DECIMALS AND

Name: _____

Date: _____

LO: I can convert between fractions, decimals and percentages and compare their values.

Converting between forms

Fractions, decimals and percentages are three ways to express the same value. Convert to the same form to compare.

Key conversions: Fraction to decimal: divide top by bottom.
Decimal to %: multiply by 100. % to fraction: put over 100 and simplify.

Equivalents

●	$\frac{1}{2} = 0.5 = 50\%$	half
●	$\frac{1}{4} = 0.25 = 25\%$	quarter
●	$\frac{3}{10} = 0.3 = 30\%$	three tenths
●	$\frac{1}{3} = 0.3 = 30\%$	0.3 repeating, not 30%
●	$0.2 = 20$ (not %)	must write 20%

Key idea

To compare, convert all values to the same form (usually decimals or percentages).

Fraction $\div$ **Decimal** $\times 100$

%

Divide then multiply to convert right

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - fraction to decimal

Convert $\frac{3}{8}$ to a decimal.

Divide numerator by denominator:

$$3 \div 8$$

$$= 0.375$$

$$\text{So } \frac{3}{8} = 0.375$$

$$= 0.375$$

Fraction means numerator divided by denominator.

Example 2 - decimal to percentage

Convert 0.45 to a percentage.

Multiply by 100

$$0.45 \times 100 = 45$$

$$= 45\%$$

$$= 45\%$$

Move the decimal point two places right.

Example 3 - percentage to fraction

Convert 35% to a fraction in simplest form.

Write as $\frac{35}{100}$

HCF of 35 and 100 is 5

$$\frac{35}{100} = \frac{7}{20}$$

$$= \frac{7}{20}$$

Always simplify by dividing by the HCF.

Example 4 - ordering

Put in order: 0.6, $\frac{5}{8}$, 62%

Convert all to decimals: $0.6, 5 \div 8 =$

$$0.625, 62\% = 0.62$$

Order: 0.6, 0.62, 0.625

$$= 0.6, 62\%, \frac{5}{8}$$

$$0.6 < 62\% < \frac{5}{8}$$

Converting to the same form makes comparison easy.

YOUR TURN – PRACTICE (deliberate practice)

1. Fractions to decimals

Convert each fraction to a decimal.

- a) $\frac{1}{4}$
- b) $\frac{3}{5}$
- c) $\frac{7}{10}$
- d) $\frac{5}{8}$

2. Decimals to percentages

Convert each decimal to a percentage.

- a) 0.35
- b) 0.8
- c) 0.05
- d) 0.125

3. Percentages to fractions

Convert to a fraction in simplest form.

- a) 40%
- b) 75%
- c) 12%
- d) 65%

4. Ordering values

Write in order from smallest to largest.

- a) 0.7, $\frac{3}{4}$, 72%
- b) $\frac{2}{5}$, 0.45, 38%
- c) 0.3, $\frac{1}{3}$, 30%
- d) 60%, 0.59, $\frac{3}{5}$

5. Mixed conversions

Complete the table: fraction, decimal, percentage.

a) $\frac{1}{5} = ?$ decimal = ? %

b) ? fraction = 0.4 = ? %

c) ? fraction = ? decimal = 85%

d) $\frac{7}{8} = ?$ decimal = ? %

e) ? fraction = 0.15 = ? %

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{1}{2} = 0.5 = 50\%$ ☐

Correct answer: _____

Reason: _____

b) $0.4 = 4\%$ ☐

Correct answer: _____

Reason: _____

c) To convert decimal to %, multiply by 100 ☐

Correct answer: _____

Reason: _____

d) $\frac{1}{3} = 33\%$ ☐

Correct answer: _____

Reason: _____

e) $75\% = \frac{3}{4}$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Three students scored $\frac{17}{20}$, 82%, and 0.84 on a test. Who scored highest? Show your conversion working.

Expression: _____

Simplified: _____

b) A shop offers $\frac{1}{3}$ off and another offers 30% off the same item priced at £60. Which is the better deal? By how much?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

